

MLDB: The Open Source Machine Learning Database *for the Cloud and for Docker*

This article talks about MLDB in relation to the cloud and Docker. It details the prominent tools for ML, describes the features of MLDB, its installation on different platforms, and the various algorithms supported by it.



Machine learning and predictive analytics are key areas of research in multiple domains including bioinformatics, computational anatomy, natural language processing, speech recognition, etc. Machine learning is used to train the software or hardware applications based on specific models for rule mining, prediction and knowledge discovery. Data scientists and analysts implement different supervised or unsupervised approaches to get accuracy and performance from raw datasets.

Machine learning (ML) and artificial intelligence (AI) are very closely related, but these terms have different perspectives. Artificial intelligence (AI) is the concept by which machines perform tasks similar to human beings, and we call such machines 'smart'.

Machine learning is the recent advanced application area of AI in which machines learn by themselves based on dynamic inputs. Machine learning based implementations are more accurate and have high optimisation.

The prominent tools for machine learning and deep learning

are MLDB, Keras, Edward, Lime, Apache Singa and Shogun. The details for each can be found on their respective websites.

Machine Learning Database (MLDB) is a powerful and high performance database system specifically developed for machine learning, knowledge discovery and predictive analytics. MLDB is FOSS and is compatible with assorted platforms. It uses the RESTful API for the storage of data, exploring the data using Structured Query Language (SQL) and, finally, it trains machine learning models.

The following are the key features of MLDB that suit a range of applications.

Speed: The training, modelling and discovery process in MLDB is highly performance-aware. It has huge processing power compared to H2O, Scikit-Learn or Spark MLlib, which are prominent machine learning libraries.

Scalability: MLDB supports vertical scaling with higher efficiency, so all memory modules as well as cores can be used simultaneously without any issues of delay or performance.

Free and open source: The community edition of MLDB